MAHATMA GANDHI UNIVERSITY , KOTTAYAM

MGU- UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

STATISTICS

MG4DSCSTA201 - Statistical Inference

Duration: 1.5 Hours **Maximum marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

Short Answer Questions

Answer any seven questions. Each question carries two marks

1. What do you mean by Law of large numbers?[K] [1]
2. Explain Unbiasedness of an estimator?[U][2]
3. Explain Efficiency of an estimator?[U][2]
4. What is Maximum likelihood method of estimation?[K][2]
5. Explain Moment method of estimation?[K][2]
6. Explain Simple and composite hypotheses ?[K][3]
7. Write hypothesis & test statistic for testing population mean has a specified value (sample size small , sample sd given) [U] [4]
8. Mention about standard error with example. [K][4]
9. Write hypothesis and test statistic for testing Chi-square tests for independence [U][5]
10. Wilcoxon signed rank test, explain?[K][5] (7*2=14)

PART B

Short Essay Questions

Answer any four questions. Each question carries six marks.

11. Explain Chebychev's inequality, write advantages & disadvantages [U][1]
12. Write Statement of Fisher-Neyman Factorization criterion[K][2]
13. Explain Method of Minimum Variance and properties of MLE? [K][2]
14. Explain critical region, significance level & power of a test ? [K] [3]

15. Survey of 320 families with 5 children each revealed the following distribution [A] [5]

No. of boys	5	4	3	2	1	0
No. of girls	0	1	2	3	4	5
No. of families	14	56	110	88	40	12

Is the result consistent with the hypothesis that male and female births are equally probable?

16. 100 students were classified according to their brilliance and community. B1 and B2 denote two levels of brilliance and A1 , A2 and A3 denote three communities. Examine whether there is any relation between community and brilliance [A] [5]

	B1	B2
A1	215	135
A2	325	175
A3	60	90

(4*6=24)

PART C

Essay Questions

Answer any one question. Each question carries 12 marks.

17. Show that if σ^2 is known $\bar{X}$ is a sufficient estimator of μ , but if μ is known S^2 is not a sufficient estimator of σ^2 in the case of a sample from N(μ , σ^2), $\bar{X}$ and S^2 being the mean and variance of the sample [A] [2]

18. The time taken by workers in performing a job by method I and method II are given below:

Method I	20	16	26	25	23	
Method II	28	33	42	35	52	34

Does the data show that the variances of time distribution by the two methods do not differ significantly? [A][4]

(1*12=12)